

Evaluating Capital Cost Estimation Programs

Five capital-cost-estimation programs are compared using a set of case studies

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Capital cost assessment is an integral part of process design when building or expanding a chemical process plant. It is required to provide project analysis and evaluation, to select among alternative designs, to plan the appropriation of funds and to serve as a basis for project cost control. Capital cost estimation is also important for an engineering student's final design project.

The accuracy of capital cost estimates depends on the available design details, the accuracy of the cost data, as well as the time available to prepare estimates. The generally accepted classification of capital cost estimates (Table 1) is published by the Association for the Advancement of Cost Engineering (AACE International). Currently, several methods and associated computer programs exist for estimating capital cost, and they provide mainly study or preliminary estimates (Table 1; Class 4).

An evaluation and comparison of the computer programs available for capital cost estimation provides a better understanding of both their use and the methods behind them. This article arose out of a systematic effort by the authors to apply and analyze several programs available for capital cost estimation. The goals of this effort were to evaluate available cost estimation programs, apply the programs to case studies of plant design and to analyze

TABLE 1. CLASSIFICATION OF CAPITAL COST ESTIMATES AACE International Recommended Practice No. 18R-97 cost estimate classification system			
Class / type	Purpose	Accuracy	Level of project completion
Class 5: Order of magnitude estimates	Initial feasibility study or screening	L: -20 to -50% H: 30 to 100%	0-2%
Class 4: Study or preliminary estimates	Concept study or feasibility	L: -15 to -30% H: 20 to 50%	1-15%
Class 3: Definitive estimates	Budget, authorization or control	L: -10 to -20% H: 10 to 30%	10-40%
Class 2: Detailed estimates	Control or bid/tender	L: -5 to -15% H: 5 to 20%	30-70%
Class 1: Check estimates	Check estimate or bid/tender	L: -3 to -10% H: 3 to 15%	50-100%

TABLE 2. PURCHASE COST (C_{PCS}) OF COMPRESSORS							
Compressor	Power, kW	Material	CapCost C_{PCS}	DFP $\Delta\%$	CCEP $\Delta\%$	EconExpert $\Delta\%$	AspenPEA $\Delta\%$
CS7-G-101	58	CS	217,350*	278%*	-4%*	-65%*	-27%
CS4-C-101	183	CS	217,350*	226%	-16%	-27%	-24%
CS5-C-101	364	CS	217,350*	454%	78%	92%	33%
CS6-C-101	3,100	CS	1,113,200	162%	128%	174%	-37%

*cost of minimum size, as size is less than the minimum size

the cost estimates for different types of equipment.

Five programs are discussed here — CapCost, EconExpert, AspenTech Process Economic Analyzer (AspenPEA), Detailed Factorial Method (DFP) and Capital Cost Estimation Program (CCEP). They were evaluated using seven case studies taken from textbooks. The selected case studies involve petroleum refining, petrochemical and biopharmaceutical processes. The cost estimates are compared and analyzed at both the equipment and plant levels to uncover the relative strengths of each of the programs used.

Estimation methods

Capital cost estimation methods in various process design reference books range from simple order-of-magnitude schemes to more detailed module costing and factorial methods. Common methods for preliminary capital cost estimation are based on the Lang factor method, the detailed factorial method and the module costing method. Most available cost estima-

tion programs in the open literature use one of these methods.

In the Lang factor method [1], plant capital cost is determined by multiplying the sum of the purchase costs of major equipment by a lumped factor called the Lang factor. Since the Lang factor is used across the whole plant, it applies the same installation factor to all equipment types in the plant, irrespective of material of construction and pressure. Sinnott and Towler [2] proposed a modified Lang factor method to estimate inside battery limit cost (C_{ISBL}) and fixed capital cost. This modified method accounts for varying installation factors in different processes, but not for different types of equipment. However, material of construction is considered through a material factor in calculating C_{ISBL} .

Another cost estimation method is the module costing method originally introduced by Guthrie [3]. This method is well accepted for estimating the cost of a new chemical plant in the preliminary stage. It is detailed, and includes a breakdown of cost categories for de-

TABLE 3. PURCHASE COST (C_{PCS}) OF HEAT EXCHANGERS

DFP ESTIMATE FOR ALL SHELL-AND-TUBE HEAT EXCHANGERS IS FOR U-TUBE TYPE; CS=CARBON STEEL; SS=STAINLESS STEEL

Heat exchanger - shell and tube (Floating-head type)			CapCost	DFP	CCEP	EconExpert	Aspen-PEA
	Area, m ²	Material	C_{PCS}	$\Delta\%$	$\Delta\%$	$\Delta\%$	$\Delta\%$
CS4-E-102	4.62	CS/CS	28,750*	-3%*	-28%*	-75%*	-37%
CS2-E-104	18.2	CS/CS	26,680	10%	-21%	-66%	50%
CS3-E-101/102/103/104	26.4	CS/CS	26,680	11%	-18%	-59%	-22%
CS4-E-103	28.2	CS/CS	26,795	12%	-17%	-58%	-24%
CS5-E-106	76.7	CS/CS	33,120	10%	-14%	-43%	-6%
CS5-E-107	127	CS/CS	40,940	8%	-15%	-38%	-3%
CS1-E-103/104	150	CS/CS	44,620	8%	-16%	-37%	-3%
CS4-E-101	405	CS/CS	87,055	11%	-26%	32%	-3%
CS5-E-101	541	CS/CS	111,320	13%	-16%	-39%	-7%
CS5-E-109	680	CS/CS	138,000	14%	-34%	-42%	11%
CS1-E-102	740	CS/CS	149,500	14%	-34%	-43%	-25%
CS5-E-108	902	CS/CS	181,700	15%	-38%	-35%	0%
CS5-E-104	2130	CS/CS	430,100	9%	-40%	-42%	-4%
CS5-E-105	2900	CS/CS	587,650	19%	-39%	-43%	-2%
CS6-E-106	11.7	SS/SS	27,945	0%	-26%	-73%	-31%
CS6-E-101	14.6	SS/SS	27,140	5%	-23%	-69%	-29%
CS4-E-106	41	SS/SS	28,060	12%	-15%	-52%	-17%
CS6-E-102	61.6	SS/SS	30,820	11%	-1%	-46%	-13%
CS6-E-105	131	SS/SS	41,515	9%	-4%	-38%	-4%
CS6-E-107	192	SS/SS	51,290	8%	-7%	-36%	-1%
CS4-E-105	269	SS/SS	63,940	9%	-21%	-36%	3%
CS5-E-102	456	SS/SS	96,025	12%	-28%	-38%	-5%
CS6-E-104	1090	SS/SS	224,250	4%	-45%	-40%	-8%
CS6-E-103	1760	SS/SS	355,350	31%	-28%	-45%	12%
CS5-E-103	2010	SS/SS	405,950	15%	-27%	-41%	-4%
Heat exchanger - shell and tube (Fixed-head type)							
CS2-E-101	14.3	CS/CS	22,540	28%	-52%	-72%	49%
CS1-E-101	20	CS/CS	26,565	9%	-19%	-64%	-22%
CS7-E-102	100	CS/SS	34,155	20%	-36%	-32%	-15%
CS7-E-103	240	CS/SS	48,990	34%	-29%	-16%	-3%
Heat exchanger - kettle reboiler							
CS2-E-102	85.3	CS/CS	120,750	-58%	-67%	-73%	67%
CS4-E-104	37.3	SS/SS	59,340	-36%	-47%	-78%	-61%
Heat exchanger - double pipe							
CS2-E-103	5.41	CS	4,796	-35%	-42%	26%	21%
Heat exchanger - plate							
CS7-E-101	57	SS	95,450	-88%	n/a*	-80%	-41%
*cost of minimum size as size is less than the minimum size							
* not available in this program and so taken from another program							

riving installed costs from purchase costs. For each piece of equipment, Guthrie provided factors to estimate the direct costs of field materials, such as piping, concrete, steel, instruments, controllers, electrical hardware, insulation and paint, as well as the direct costs of field labor used for their installation. These factors include material erection and equipment setting, as well as the indirect costs involved in installation, such as insurance, construction overhead and contractor engineering expense. The bare module cost (C_{BM}) for a given piece of equipment i is thus defined as:

$$C_{BM,i} = C_{DE,i} + C_{IDE,i} = [C_P^o + C_M + C_L + C_{FIT} + C_O + C_E]_i \quad (1)$$

where $C_{BM,i}$, $C_{DE,i}$, $C_{IDE,i}$ and C_P^o are, respectively, bare module cost, direct cost, indirect cost and purchase cost of equipment i in base conditions (that is,

carbon steel material and atmospheric pressure). C_M is the cost of field materials required for installation, C_L is the cost of labor to install equipment and materials, C_{FIT} is the cost of freight, insurance and taxes, C_O is the cost of construction overhead and C_E is the cost of contractor engineering expenses. For the whole plant, Ulrich and Vasudevan [4] propose to find the total module cost (C_{TM}) by multiplying $C_{BM,i}$ for all equipment types by the factor 1.18 to account for contingency and contractor fee.

$$C_{TM} = \sum (C_{BM,i} + C_{Cont,i} + C_{Fee,i}) = 1.18 \sum C_{BM,i} \quad (2)$$

where $C_{Cont,i}$ is contingency fee and $C_{Fee,i}$ is the contractor fee.

Computer programs

The five programs used in this study are CapCost, DFP, CCEP, EconExpert and AspenPEA. Most programs use a para-

metric-cost model for the cost estimates, which is useful in early conceptual estimates [5]. The cost equation constants used in these programs are obtained from vendor quotes or from past literature data. As discussed by Woods [6] and Walas [7], when cost data are assembled from vendor quotes, they exhibit scatter due to different qualities of equipment design, fabrication, market conditions, vendor profit and other considerations. Hence, the accuracy of published equipment-cost data may be no better than $\pm 25\%$, and therefore the estimating method based on these data can only be used for study or preliminary estimates.

CapCost is available with the process design book of Turton and coauthors [1]. It is based on the module costing method, written in Visual Basic, and can be used for estimating preliminary process cost. Bare module cost (C_{BM}) is defined as the sum of the direct and indirect expenses for purchasing and installing equipment; the total module cost (C_{TM}) is defined as the sum of the bare module cost, contingency and fee; and the grassroots plant cost (C_{GR}) is defined as the sum of the total module cost and the auxiliary facilities costs. To estimate the bare module cost and purchase cost of equipment, Turton and colleagues proposed the following:

$$C_{BM} = C_P^o \times F_{BM} = C_P^o (B_1 + B_2 F_M F_P) \quad (3)$$

$$\log C_P^o = K_1 + K_2 \log(S) + K_3 [\log(S)]^2 \quad (4)$$

where S represents a parameter for the equipment size or capacity. Values for the constants B_1 and B_2 , equipment-specific constants K_1 , K_2 and K_3 , as well as correlations and plots for F_{BM} , F_M , F_P and C_P^o of different equipment can be found in the appendices in Ref. 1. These data are based on surveys of equipment vendors between May and September 2001. The *Chemical Engineering Plant Cost Index* (CEPCI; see p. 56 and www.che.com/pci for current value) value of 397 for this period can be used for escalating cost to a different time.

Detailed Factorial Program (DFP) is based on the detailed factorial estimates method described in Ref. 2. For this program, the purchase cost, C_P^o , of the major equipment items is esti-

TABLE 4. PURCHASE COST (C_{PCS}) OF MIXERS

Mixer Details			CapCost	DFP	CCEP	EconExpert	AspenPEA
	Power, kW	Material	C_{PCS}	$\Delta\%$	$\Delta\%$	$\Delta\%$	$\Delta\%$
CS3-M-101	0.7	CS	37,375*	38%*	n/a*	-83%*	-70%*
CS7-M-105	0.015	SS	37,375*	-94%	n/a*	-99%	-97%
CS7-M-104	2	SS	37,375*	-37%	n/a*	-81%	-67%
CS7-M-103B	2.5	SS	37,375*	-37%	n/a*	-79%	-64%
CS7-M-103A	4.2	SS	37,375*	-37%	n/a*	-74%	-57%
CS7-M-102	7	SS	40,595	-36%	n/a*	-68%	-52%
CS7-M-101	146	SS	103,730	16%	n/a*	56%	-13%

*cost of minimum size as size is less than the minimum size

*mixer cost not available for CCEP

mated using the following:

$$C_P^o = a + bS^n \quad (5)$$

Cost constants a and b , available in Ref. 2 for different equipment items, are mainly for carbon-steel material, and correspond to January 2007 (CEPCI of 509.7). The ISBL cost and fixed capital cost are calculated according to the method in Ref. 2.

Capital Cost Estimation Program (CCEP) uses cost correlations in Seider [8] for estimation of free-on-board purchase cost of equipment. The material factor and Guthrie's bare module factor are used thereafter to estimate the installed cost of that equipment. Seider [8] developed the purchase cost correlations for common process equipment, based on available literature sources and vendor data. A list of these cost correlations can be found in Ref. 8, using CEPCI = 500. The purchase cost of the major equipment items is estimated using the following:

$$C_P^o = e^{[A_0 + A_1[\ln(S) + A_2[\ln(S)]^2 + \dots]} \quad (6)$$

Values of constants A_0 , A_1 and A_2 for various equipment items can be found in Ref. 8. In addition to size, other factors, such as material of construction and operating pressure, are also taken into consideration in estimating the purchase cost. Therefore, the cost equations are often in the following form:

$$C_P = F_M F_P C_P^o \quad (7)$$

where C_P is the purchase cost of equipment; F_M is the material factor and F_P is the pressure factor. CCEP and DFP were developed in Microsoft Excel and Visual Basic environments, by Wong [9] and Huang [10], respectively, as part of research projects supervised by the second author (these programs can be obtained from the authors).

EconExpert is a Web-based interactive software for capital cost estimation [11]. Similar to CapCost, the equipment module costing method is used to calculate bare module cost and total module cost from the purchase cost of equipment. The purchase cost data and bare module factors used can be found in Ref. 4. In this textbook, the cost data are expressed in graphical form, whereas in EconExpert, the

plots are represented as polynomial equations for calculation of the purchase cost. Multiple regression is used to fit the data if the purchase cost is dependent on more than one variable. The cost data and correlations in EconExpert are for a CEPCI of 400 [4].

AspenPEA is built on Aspen Icarus technology, and is designed to generate both conceptual and detailed estimates [12]. It takes a unique approach, representing equipment by comprehensive design-based installation models. AspenPEA claims to contain time-proven, field-tested, industry-standard cost modeling and scheduling methods [12]. AspenPEA Version 7.1 was used for this study (this version follows 2008 data, where the CEPCI is 575).

Equipment examples

Seven case studies based on process design data available in Refs. 1, 4, 8 and 13 were chosen for this study. In Case 1 from Ref. 13, a petroleum refinery distillation column (the alkylate splitter) is employed to separate a mixture of C4 to C14 hydrocarbons into two streams. Case 2 is the monochlorobenzene (MCB) separation process in Ref. 8. Case 3 is a crystallization process from the same source. Case 4 is the formalin production process from methanol using silver catalyst [1]. A styrene production process in the same reference is Case 5. Case 6 is an alternative synthesis of maleic anhydride via benzene in a shell-and-tube reactor with catalyst [1]. Case 7 is a β -galactosidase batch process via recombinant *Escherichia coli* [4].

The five computer programs were used to evaluate capital costs for each case. A CEPCI of 575 was used throughout to allow cost estimates by all programs to be compared meaningfully. The CEPCI generally trends upward over time. An exception was 2009, when the CEPCI dropped to 525 (from 575 in 2008) so, the CEPCI seems likely to return to 575 in the near future.

The CapCost program is used as the reference for comparison, although any

other program could have been used. The base condition (equipment made of carbon steel and operating near ambient pressures) purchase cost (C_{PCS}) and total module cost (C_{TM}) are compared and analyzed. In DFP, CCEP and AspenPEA, calculated purchase cost is directly based on actual material of construction and pressure. Hence, C_{PCS} in these programs is calculated by entering equipment details for the same condition as the one required using carbon steel as the material of construction and 1 barg as the operating pressure.

As the module costing technique is not used in DFP and AspenPEA, only C_{ISBL} and C_{Direct} , respectively, are available from these programs. By analyzing the cost factors, including C_{TM} , C_{ISBL} and C_{Direct} , it is observed that C_{TM} includes engineering design, contract fees and contingency fees, while the other two do not. Hence, C_{TM} in DFP and AspenPEA is calculated by Equations (8) and (9), respectively.

$$C_{TM} = C_{ISBL} (1 + DE + X) \quad (8)$$

$$C_{TM} = C_{Direct} (1 + DE + X) \quad (9)$$

Here, C_{ISBL} is the inside battery limit cost; and $DE = 0.3$ and $X = 0.1$, according to the factors stated in the detailed factorial method [2].

Some process equipment may have sizes outside the valid range for the cost correlations or for the program. For all programs, if a piece of equipment has a size above the valid range, it is divided into multiple units of smaller size and the costing is done by summing up C_{TM} of multiple smaller units. If equipment has a specification below the valid range, then its cost is estimated by taking the lower limit of the valid range. Similarly, for flowrates less than the minimum amount required for costing, the minimum flowrate is used instead of the actual flowrate. This causes some uncertainty in the predicted cost.

In the following sections, base condition purchase cost (C_{PCS}) and total module cost (C_{TM}) for equipment in the seven case studies are compared.

TABLE 5. PURCHASE COST (C_{PCS}) OF PUMPS

Centrifugal pumps			CapCost	DFP	CCEP	Econ-Expert	Aspen-PEA
	Power, kW	Material	C_{PCS}	$\Delta\%$	$\Delta\%$	$\Delta\%$	$\Delta\%$
CS4-P-101	0.3	CS	3,554*	152%	50%*	7%	55%
CS5-P-104	0.38	CS	3,554*	156%*	49%*	14%*	57%*
CS6-P-101	0.46	CS	3,554*	152%*	62%*	21%	57%
CS5-P-105	0.75	CS	3,554	165%	46%	39%	60%
CS2-P-102	1.12	CS	3,577	164%	48%*	57%	35%
CS1-P-102	1.7	CS	3,726	165%	27%*	73%	53%
CS4-P-102	1.7	CS	3,726	110%	46%*	73%	53%
CS5-P-102/103	2	CS	3,807	166%	54%*	79%	50%
CS2-P-101	2.24	CS	3,876	176%	28%*	83%	37%
CS5-P-106	2.65	CS	3,979	209%	53%	89%	76%
CS1-P-101	2.8	CS	4,025	251%	26%	91%	89%
CS6-P-102	5.84	CS	4,807	297%	120%	110%	123%
CS5-P-101	6.4	CS	4,934	289%	75%	112%	107%
CS1-P-103	24	CS	8,257	204%	24%	117%	10%
CS4-P-103	0.5	SS	3,554*	146%	59%*	24%	57%
CS6-P-105	1.08	SS	3,565	143%	223%	56%	57%
CS7-P-101	2.3	SS	3,887	163%	51%*	84%	50%
CS6-P-106	3.69	SS	4,267	145%	294%	99%	132%
CS6-P-104	10.4	SS	5,808	438%	140%	118%	58%
Reciprocating pumps							
CS3-P-104	0.2	CS	7,659	n/a*	-25%*	11%	-9%*
CS3-P-105	0.24	CS	7,613	n/a*	-24%*	11%	-8%*
CS3-P-103	0.6	CS	9,258	n/a*	-55%	22%	-24%*
CS3-P-101	22.4	CS	47,380	n/a*	-21%	22%	-38%*
CS3-P-102	29.6	CS	57,500	n/a*	-35%	22%	-49%*
CS6-P-103	0.16	SS	7,176	n/a*	-29%*	4%	-3%*
Diaphragm pumps							
CS7-P-105	1	SS	10,730	n/a*	45%*	26%	-50%
CS7-P-103	1.9	SS	13,455	n/a*	182%*	27%	-50%
CS7-P-102	3.7	SS	17,710	n/a*	-36%	27%	-51%
CS7-P-104	60	SS	95,105	n/a*	17%	25%	-87%

* cost of minimum size as size is less than the minimum size

+ not available in this program and so taken from another program

Compressors

Major compressor types are trunk-piston and crosshead reciprocating compressors, diaphragm compressors, centrifugal compressors and axial compressors. Case studies 4–7 involve compressors, and purchase costs of these compressors at base conditions are presented in Table 2. Total module cost is shown in online table I.

In these and subsequent tables, the cost estimate by CapCost is given, and the cost estimate by other methods is expressed as the percent difference from that by CapCost, such that $\Delta = 100 \times (\text{cost using other method} - \text{cost using CapCost}) / \text{cost using CapCost}$. Hence, a positive (or negative) Δ value means the cost estimate by that method is more (or less) than that determined by CapCost.

Results in Table 2 indicate that compressor purchase cost estimates by CapCost, CCEP and AspenPEA are comparable, while the purchase costs given by EconExpert increase very fast with increasing size. In fact, DFP gives a very high cost compared to all other programs, which is mainly due to the cost equation given in Ref. 2. From

this equation, even the lower limit (power of 75 kW) gives a cost as high as \$714,050. Such a high cost may be due to the data source for calculation of empirical constants. Differences in total module cost of compressors by different programs are larger than those in purchase costs (Table online), which is due to differences in installation and other factors used for C_{TM} .

Heat exchangers

Common heat-exchanger types are shell-and-tube, double-pipe, air-cooled fin fan, and compact heat exchangers, including plate-and-frame and spiral plate types. Because heat exchangers are common, the data available from vendors and other sources are voluminous. The wealth of data results in relatively more-accurate cost estimation. For example, the purchase cost and total module costs of floating-head and fixed-head heat exchangers predicted by all programs are comparable, with somewhat lower cost by CCEP and EconExpert (Table 3). For kettle reboilers, CapCost predicts 50 to 80% higher purchase and total module costs than the other four pro-

grams. For double-pipe heat exchangers, all programs predict comparable purchase cost. However, AspenPEA predicted a total module cost three times that by CapCost (online table II). The most significant factor contributing to the high total module cost in AspenPEA is the piping cost. While other programs calculate piping cost as a small fraction of purchase cost, AspenPEA model calculates piping cost based on respective equipment. For double-pipe heat exchangers, the piping cost is around three times the purchase cost, which contributes $10 \times C_{PCS}$ to C_{TM} .

Figure 1 shows the total module cost of floating-head heat exchangers calculated in all five programs as a function of heat transfer area. Since only the U-tube shell-and-tube heat exchanger cost equation is given in Ref. 2, the DFP estimate in Figure 1 is for this type. Since floating-head heat exchangers are generally costlier than the U-tube type, the floating-head heat exchanger cost by DFP is expected to be slightly higher than that in Figure 1.

For smaller heat transfer areas (up to 400 m²), all five programs predict total module cost for floating-head exchangers with less than 50% deviation using the CapCost value as a reference. As the area increases, the deviations become more significant. In the seven case studies, many of the floating-head heat exchangers have areas below 400 m². Hence, all five programs predicted similar costs. However, when the heat transfer area is large, CapCost and DFP predict a relatively higher cost than the other programs.

Figure 2 shows the variation in the total module cost of fixed-head heat exchangers calculated by all five programs, with heat transfer area. Since only the U-tube shell-and-tube heat exchanger cost equation is given in Ref. [2], the DFP estimate in Figure 1 is for this type. CCEP and EconExpert predict lower costs than the other three programs for all sizes of fixed-head heat exchanger. However, for smaller areas, (up to 300 m²) the percent deviation is relatively small. In the case studies, most of the fixed-head heat exchangers have areas below 300 m². Hence, all five programs predicted a similar cost. As the area

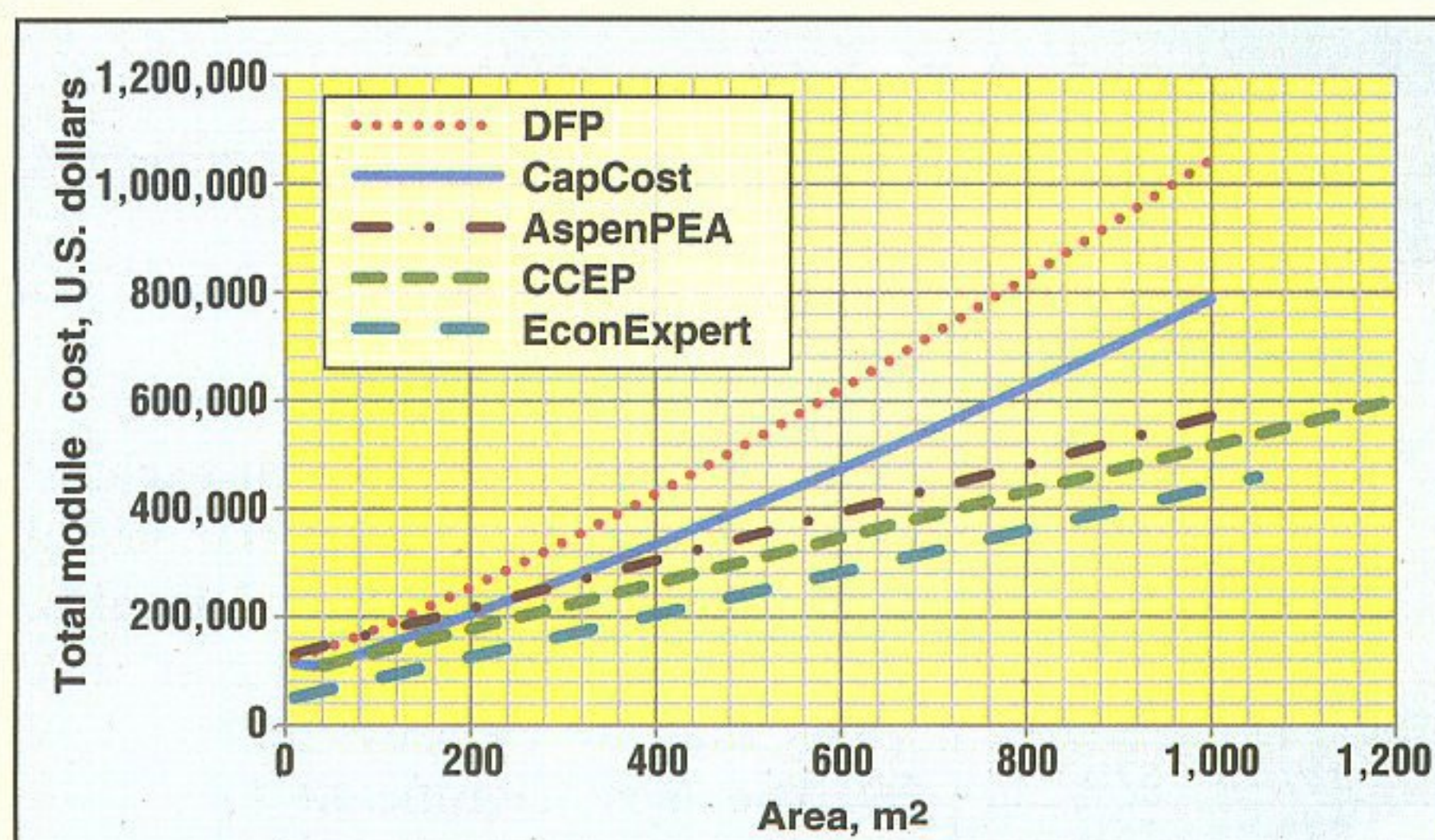


FIGURE 1. As the area of floating-head heat exchangers increases the deviation in total module cost estimates grows

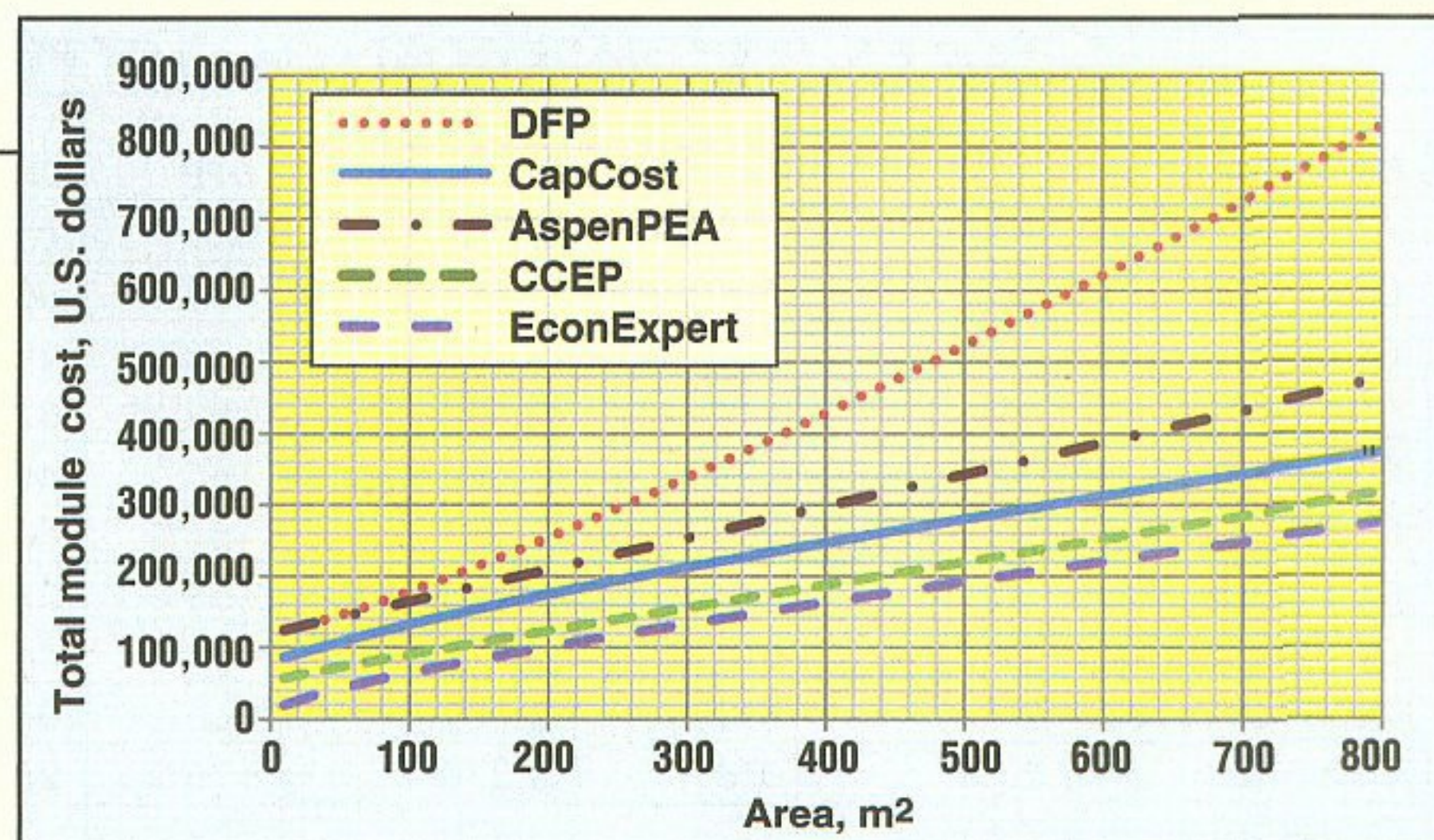


FIGURE 2. Fixed-head heat exchangers are generally less expensive, but deviations in estimates still increases with area

increases, the differences become more significant, with DFP predicting a significantly higher cost than all the others. This may be partially due to the mapping of fixed-head heat exchanger to U-tube heat exchanger in DFP. In general, fixed-head heat exchangers are slightly less expensive than U-tube heat exchangers [8] (Table 3, Figures 1–2 and online table II).

The total module cost of double-pipe heat exchangers is plotted against heat transfer area in Figure 3. AspenPEA predicts a significantly higher cost than all other programs (again, mainly due to the higher cost of piping predicted). For double-pipe exchangers, EconExpert and CapCost predicted similar costs, while DFP and CCEP predicted slightly lower cost (Figure 3).

Mixers

Of the huge variety in mixer designs available, common types are ribbon and tumbler mixers, kneaders and mullers. Mixer costing is not included in the CCEP program. Mixer purchase costs differ by 50–100% among different programs (Table 4). The deviation in total module cost (online table III) is similar to that for the purchase cost for EconExpert and AspenPEA. However, the total module cost given by DFP is higher due to a large installation factor for mixers in this method. The cost estimate by DFP and EconExpert increases faster than that by CapCost, and so the former two predict significantly higher cost than CapCost for large mixers. For example, the total module cost of CS7-M-101 by DFP and EconExpert are, respectively, 193% and 160% of that by CapCost, even though the purchase costs are comparable. This is due to the limitation of mixer size in DFP, which requires its costs to be estimated as that of two smaller mixers. Such a mapping increases the cost significantly. Also, selection of ma-

TABLE 6. PURCHASE COST (C_{PCs}) OF TOWERS								
Tray Towers				CapCost	DFP	CCEP	EconExpert	AspenPEA
	Dia., m	Length, m	Material	C_{PCs}	$\Delta\%$	$\Delta\%$	$\Delta\%$	$\Delta\%$
CS2-T-102	0.5	12.8	CS	26,105	-9%	249%*	n/a*	0%
CS2-T-101	0.9	21.9	CS	63,135	-19%	121%	n/a*	-4%
CS1-T-101	3.0	30.0	CS	483,000	26%	29%	n/a*	76%
CS5-T-101	3.6	28.0	CS	702,650	6%	16%	n/a*	-18%
CS6-T-102	1.1	18.0	SS	78,775	25%	89%	n/a*	87%
CS4-T-102	2.5	19.0	SS	230,000	24%	48%	n/a*	21%
CS6-T-101	4.2	10.0	SS	565,800	-5%	-57%	n/a*	-51%
Packed Tower								
CS4-T-101	0.9	10.0	CS	21,045	63%	284%*	n/a*	135%

* cost of minimum size as size is less than the minimum size

+ cost trays and packing are not included in EconExpert, and hence not compared

terials of construction is not available in CapCost, which results in similar costs for mixers of different materials in that program.

Pumps

Commonly used pumps in chemical processing plants are radial-centrifugal, plunger-reciprocating, diaphragm and external-rotator-gear pumps. For centrifugal pumps, the four other programs predict higher purchase and total module costs than CapCost, with DFP giving costs up to three times of those predicted by CapCost (Table 5). One reason for this difference may be due to using different input data. While CapCost and EconExpert programs require only pump power for cost prediction, DFP requires flowrate and liquid density, in addition to pump power, and CCEP and AspenPEA require flowrate and pump head. Hence, two pumps with the same power, but different flowrates due to design pressures, will have the same cost in CapCost and EconExpert, but different costs in the other three programs. However, even though both CapCost and EconExpert use pump power to predict the cost, they have rather different results. This difference is due to the variation in cost data used in these two programs to calculate purchase cost.

For centrifugal pumps, purchase and total module costs by AspenPEA deviate by 50–100% and 100–400%, respectively, from those of CapCost (Table 5). The high total module cost is mainly due to the higher piping cost given by AspenPEA. Reciprocating pumps are not available in DFP. The other four programs give similar purchase costs (Table 5 and online table). However, CCEP predicts 50% lower total module cost than CapCost (online table). This is due to different module factors used in the two programs. For diaphragm pumps with small power, the deviation in total module cost between different programs is within 70%. However, when the power is large, the deviation between different programs increases further (online table IV).

In Figure 4, the total module costs of centrifugal pumps are plotted against their power. In general, DFP and AspenPEA predict higher cost than the others. For low-power pumps, DFP and AspenPEA predict similar costs. However, as the power increases, DFP predicts a higher cost than AspenPEA. CapCost gives the lowest, and EconExpert predicts a cost in the middle range. This observation coincides with the results obtained in case studies above. Compared to other programs, the cost

TABLE 7. PURCHASE COST (C_{PCS}) OF VESSELS
All vessel pressures less than 10 barg except CS1-V-101 with pressure = 60 barg

Vessel - horizontal				CapCost	DFP	CCEP	EconExpert	Aspen-PEA
	Dia., m	Length, m	Material	C_{PCS}	$\Delta\%$	$\Delta\%$	$\Delta\%$	$\Delta\%$
CS2-V-101	0.9	1.4	CS	5,014	155%*	372%*	1%	21%*
CS6-V-101	1.2	3.5	CS	9,212	308%*	188%*	1%	89%
CS5-V-102/103	1.3	3.9	CS	10,603	52%	183%	2%	105%
CS1-V-102	1.6	4.7	CS	14,490	38%	87%	4%	78%
CS1-V-101	2.7	8.0	CS	38,640	-10%	93%	-4%	14%
CS6-V-103	1.3	3.9	SS	10,787	552%*	180%*	2%	104%
CS4-V-101	1.7	6.9	SS	9,718	173%	345%	5%	111%
CS7-VS-105	2.0	3.2	SS	15,295	44%	189%	27%	80%
CS7-VS-106	2.5	6.1	SS	29,670	19%	135%	30%	43%
CS6-V-102	4.4	13.2	SS	136,850	260%	-14%	-45%	-28%
Vessel - vertical								
CS2-V-102	0.9	3.7	CS	6,992	629%	441%*	102%	40%*
CS3-V-101	1.8	3.7	CS	15,985	12%	191%	58%	72%
CS5-V-101	2.5	7.4	CS	40,135	-14%	130%	35%	17%
CS5-V-104	4.0	34.5	CS	386,400	-56%	-13%	-31%	-27%
CS7-V-102	0.4	6.0	SS	4,025	291%	923%*	276%	151%
CS7-VS-102	0.5	1.0	SS	2,841	430%*	1301%*	71%*	127%
CS7-V-101	0.5	5.0	SS	4,761	238%	765%*	194%	144%
CS7-VS-104	1.2	1.8	SS	6,348	141%	527%	11%	56%
CS7-VS-103	1.5	2.0	SS	8,591	93%	414%	17%	26%
CS7-VS-101	3.5	10.5	SS	97,405	-44%	85%	-15%	-54%
Vessel - jacketed								
CS7-VS-110	2.0	3.8	SS	59,300+	n/a*	n/a*	n/a*	n/a*
CS7-VS-109	2.3	4.5	SS	75,600+	n/a*	n/a*	n/a*	n/a*
CS7-VS-108	3.0	4.8	SS	124,500+	n/a*	n/a*	n/a*	n/a*
CS7-VS-107	3.2	8.5	SS	199,100+	n/a*	n/a*	n/a*	n/a*

*cost of minimum size as size is less than the minimum size

*not available in this program and so taken from another program

of a centrifugal pump in CCEP depends on an additional factor: the stages and split-case orientation. For lower-power pumps (up to 55 kW and flowrates up to 57 L/s), the pump is operating in one stage and vertical split case, and the cost is lower. For higher power and flowrates, the pump operates in two or more stages and horizontal split case, and the cost is higher. Hence, a jump is observed for the CCEP cost in Figure 4 due to the change of stages. The CCEP program warns the user if the pump is not operating in the appropriate stage.

When C_{TM} for reciprocating pumps is plotted against shaft power, EconExpert gives the highest cost of the four programs, followed by CapCost and CCEP, while AspenPEA predicts the lowest (reciprocating pumps are not available in DFP). This result is similar to what is observed in case studies.

Towers

Towers are vertical-pressure vessels for separation operations, such as absorption, distillation and stripping. They contain trays or packing, plus manholes and nozzles. For both towers and vessels, shell thickness is required for cost estimation in DFP. Since in most cases, the designed pressure is provided in the equipment data, the

following equation from Ref. 1 is used to estimate the shell thickness.

$$t = [PD \div (2S_{max} E - 1.2P) + CA] \quad (10)$$

Here, t is the shell thickness in meters, P is the design pressure in bars, D is the diameter of the vessel (meters), S_{max} is the maximum allowable working pressure of the material (bars), E is the weld efficiency and CA is the corrosion allowance (0.0035 m).

For tray towers, the purchase cost given in EconExpert does not include trays and packing, while those are included in the total module cost. Hence, EconExpert's purchase cost is not compared with CapCost (Table 6). DFP gives a purchase cost similar to CapCost, whereas CCEP is higher when the diameter is small (or lower when large). When the total module cost of the tower is compared, for towers of large diameter, all programs give similar results (online table V). CCEP shows a relatively higher total module cost (+59%). For small-dia. towers, CapCost predicts a much lower cost than the others. A packed tower is also evaluated (Table 6). CapCost predicts a much lower cost than the rest. Both CCEP and AspenPEA predict the cost of this packed tower to be significantly higher than that by CapCost.

Vessels

Vessels are used in chemical processing plants as reflux drums, flash drums, knock-out drums, settlers, chemical reactors, mixing vessels and storage drums. In general, for horizontal vessels with low design pressure and small diameter, CapCost and EconExpert predict similar purchase and total module costs while the others give much higher costs (Table 7). For horizontal vessels of large diameter, CapCost, DFP and EconExpert give similar results, while CCEP predicts a higher cost. The total module cost in AspenPEA is exceptionally high due to the high instrumentation cost, which is three times C_{PCS} . This contributes to C_{TM} as ten times C_{PCS} (online table VI).

For vertical vessels, CapCost gives significantly lower costs than the other four programs. However, when the design pressure of the vessel is very high (CS1-V-101), CapCost predicts a very high pressure factor according to:

$$\log_{10} F_P = C_1 + C_2 \log_{10} P + C_3 (\log_{10} P)^2 \quad (11)$$

where C_1 , C_2 and C_3 are constants that can be found in Ref. 1. Hence, although the purchase cost from CapCost is comparable or even lower than in other programs, the total module cost in CapCost is much higher than that by the other four programs.

The plot of total module cost of horizontal vessels against volume (Figure 5) shows that CapCost and EconExpert predict similar total module costs for horizontal vessels. DFP predicts a slightly higher cost, whereas AspenPEA and CCEP predict the highest costs. Also, CCEP's cost of horizontal vessels increases faster with increasing volume than that by AspenPEA. When volume is large, CCEP predicts a much higher cost than AspenPEA, and the deviation becomes more significant as the volume and size increase. The horizontal vessels evaluated in the case studies are all below 100 m³ (Table 7), and their costs follow these trends.

For vertical vessels of small diameter, CapCost predicts lower purchase and total module costs than the others (Figure 6, Table 7 and online table VI). However, for large-diameter vertical vessels (CS7-VS-101), the cost given

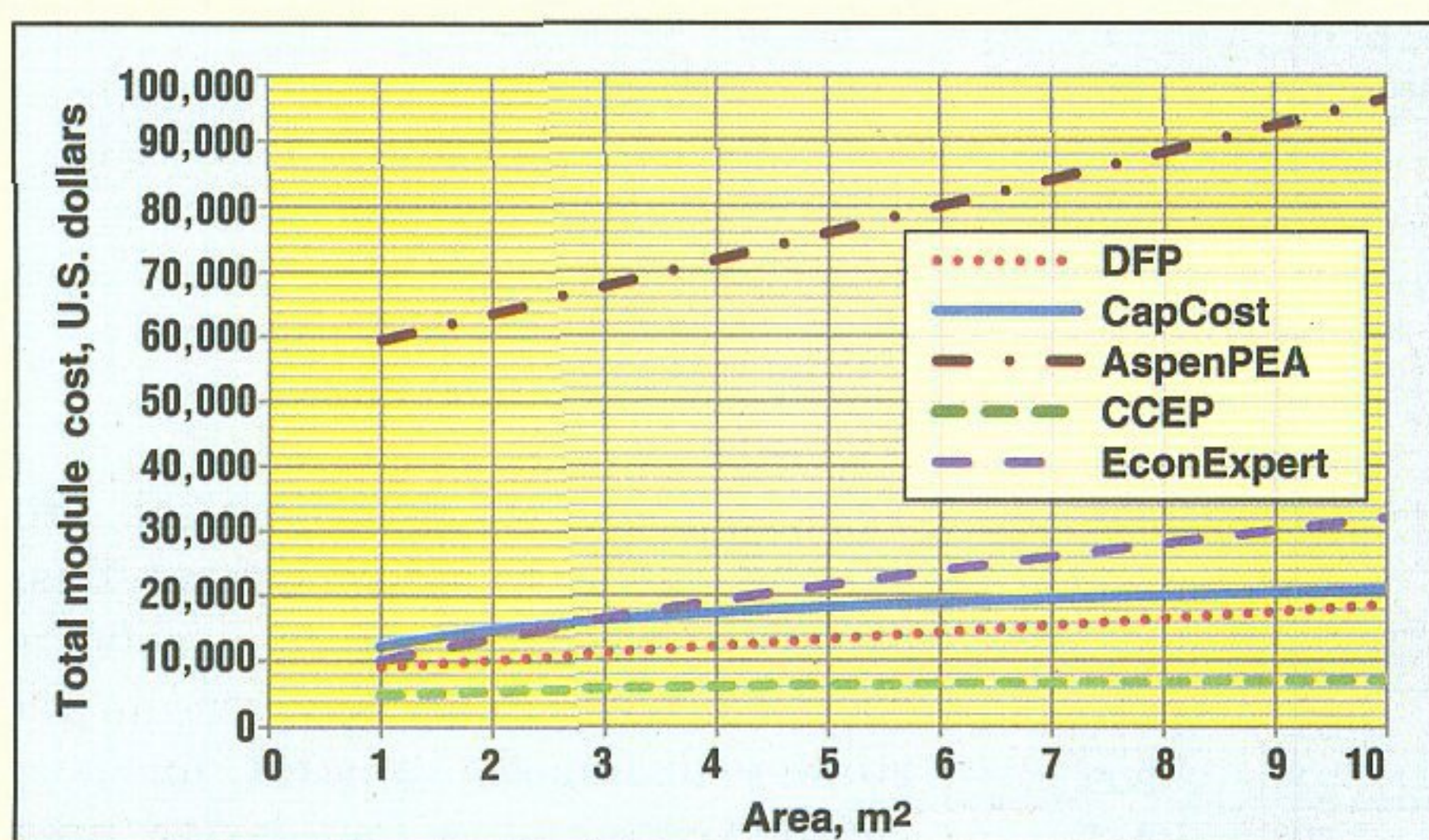


FIGURE 3. Due to its treatment of piping, DFP predicts a higher total module cost for double-pipe heat exchangers

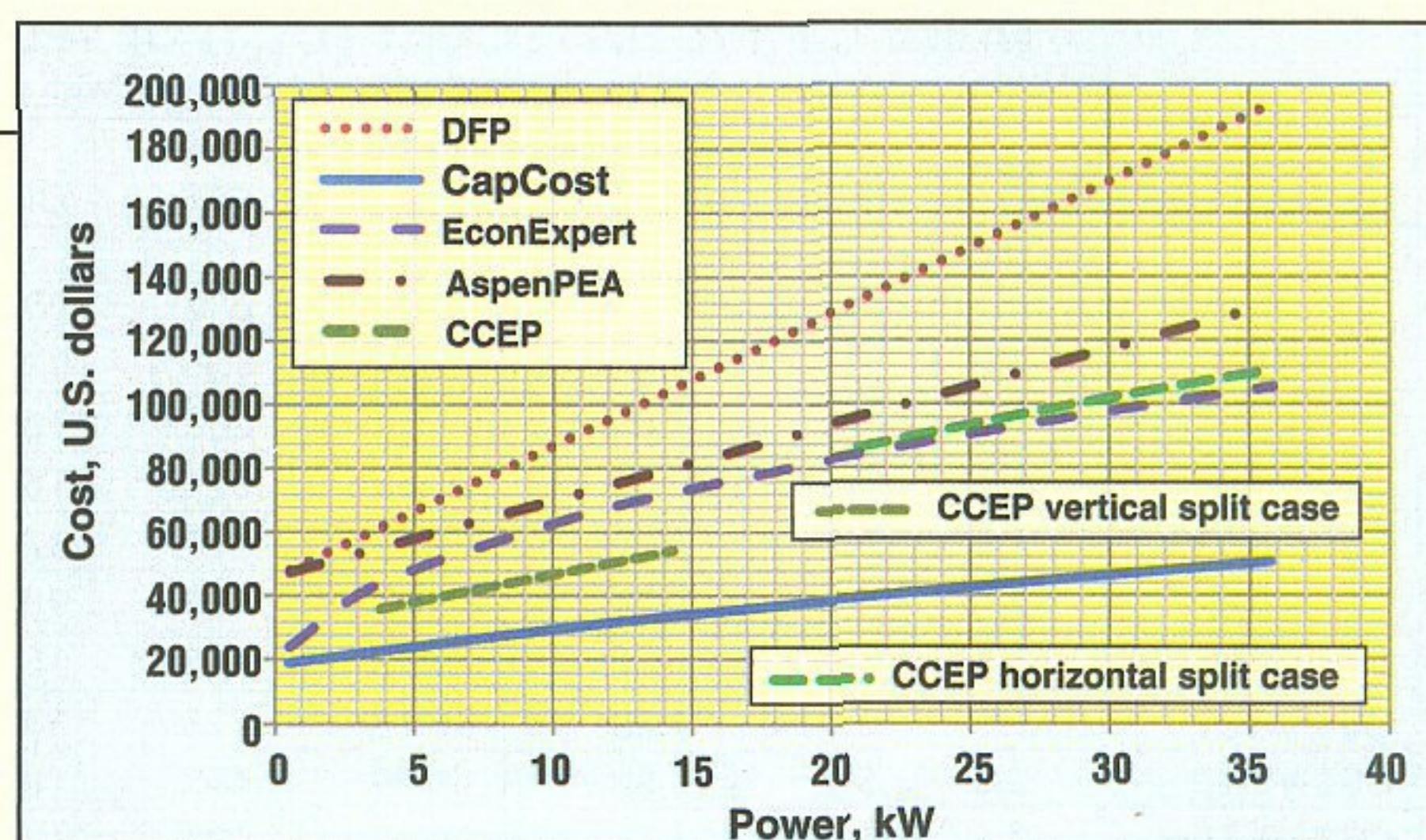


FIGURE 4. DFP and AspenPEA predict higher total module costs of centrifugal pumps than other programs

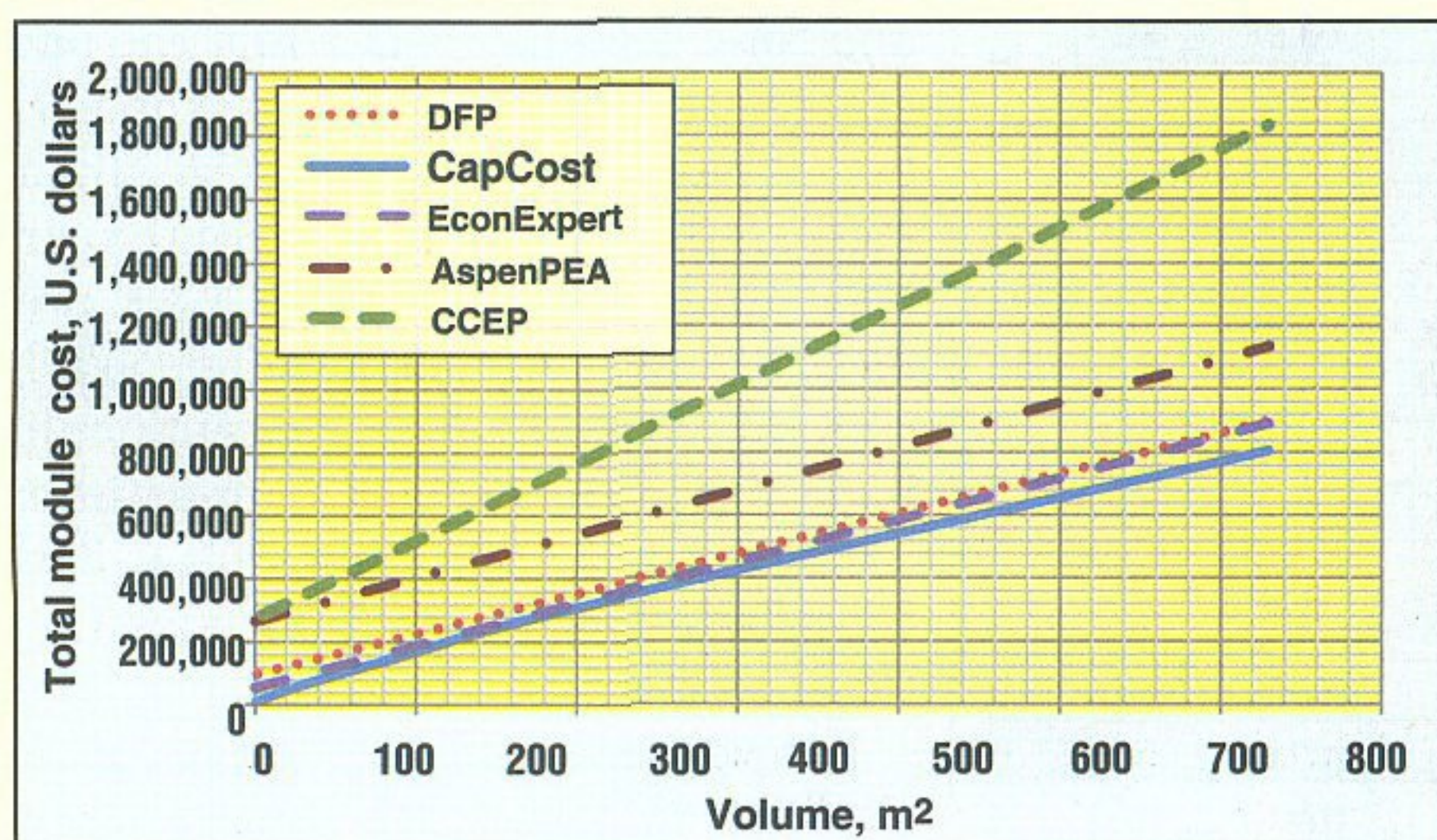


FIGURE 5. Total module cost predictions for horizontal vessels are lowest for CapCost

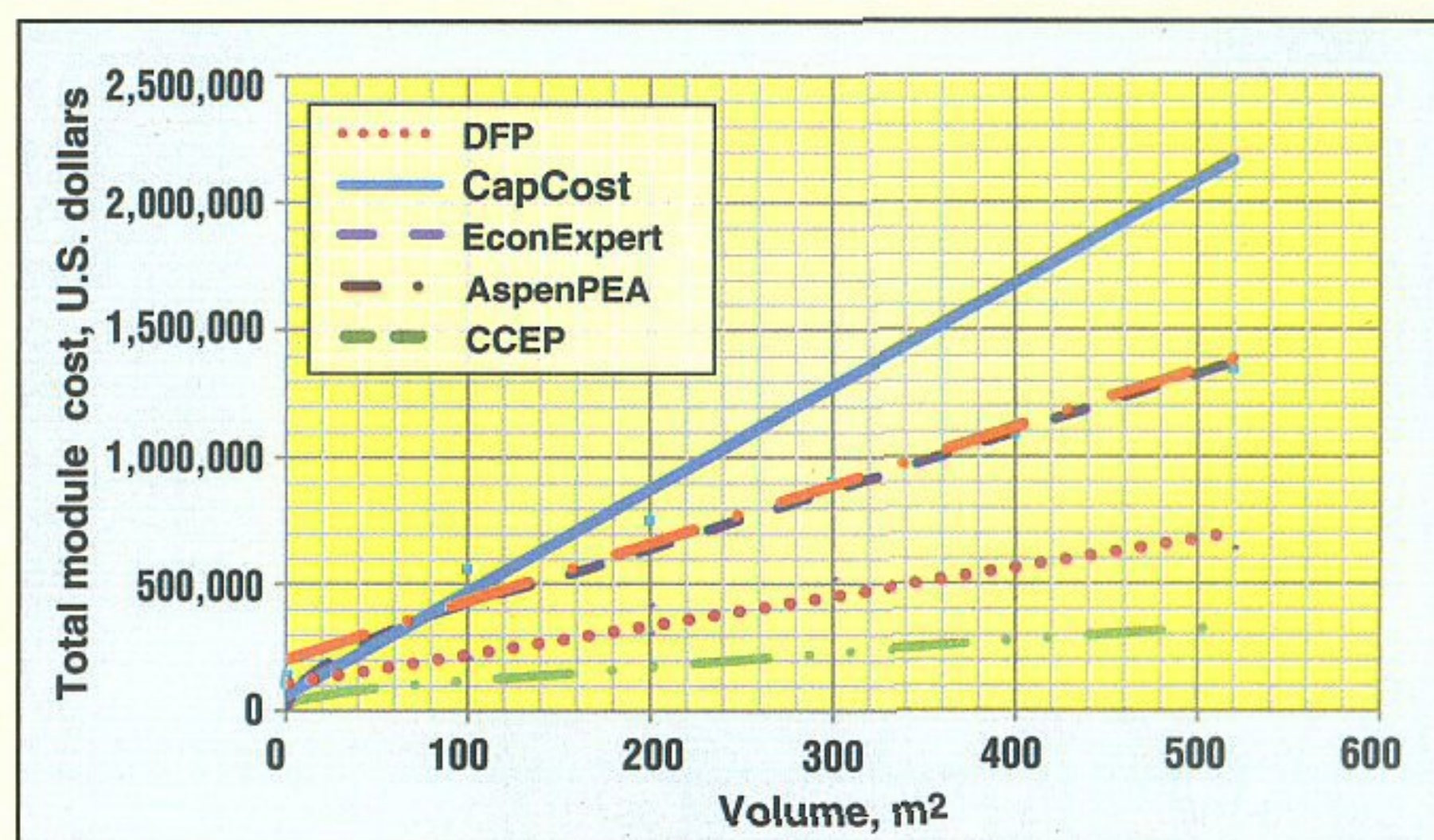


FIGURE 6. The rate of increase in predicted total module cost for vertical vessels is greatest with CapCost

by CapCost is lower than CCEP only. The rate of cost increase by CapCost is faster than that by other programs. While CapCost's cost is low for small volumes, its predicted cost is the highest for volumes more than 200 m³.

Other equipment

In this section, other pieces of equipment, such as centrifuges, crystallizers, fired heaters, mills, filters and jacketed and agitated reactors are discussed (Table 8). Crystallizers and centrifuges are not listed in DFP and EconExpert. AspenPEA gives high purchase and total module costs compared to CapCost and CCEP for both crystallizers and centrifuges. The de-

viation is mainly due to differences in the C_{PCS} predicted based on different correlations in the programs.

In case studies five and six, fired heater cost is evaluated. For CS5-H-101, all programs give similar purchase and total module costs with DFP giving a relatively higher cost than the rest. However, the cost of CS6-H-101 shows significant deviation for different programs. Due to limited data, the trend of fired heater cost is not clear.

In case study seven, there are equipment types that are not present in earlier case studies. The first is a mill, whose costing is available only in EconExpert and AspenPEA. Comparing the results in Table 8 (and online table VII),

both purchase and total module costs predicted by EconExpert are half that predicted by AspenPEA. This is mainly caused by the different approach in calculating the purchase cost of the mill in different programs. Filters and filtering centrifuges can only be found explicitly in CapCost, EconExpert and AspenPEA. In AspenPEA, there are more types of filters and centrifuges to which to map the process equipment, compared to fewer choices for CapCost and EconExpert. Due to the mapping to different type of filters, the costs predicted deviate from one another. However, the deviation is within the acceptable range for those which are within the size range. Furthermore, jacketed

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TABLE 8. PURCHASE COST (C_{PCS}) OF OTHER TYPES OF EQUIPMENT

Centrifuges			CapCost	DFP	CCEP	EconExpert	AspenPEA
	Dia., m	Material	C_{PCS}	$\Delta\%$	$\Delta\%$	$\Delta\%$	$\Delta\%$
CS3-Ct-101	2	CS	303,600	n/a ⁺	n/a ⁺	n/a ⁺	49%
CS3-Ct-102	2	CS	303,600	n/a ⁺	n/a ⁺	n/a ⁺	49%
Crystallizers							
CS3-Cr-101	21.9	CS	139,150	n/a ⁺	-76%	n/a ⁺	210%
Fired heater							
CS6-H-101	26,800	CS	1,078,700	63%	112%	-46%	59%
CS5-H-101	360,000	CS	6,405,500	-23%	-15%	-13%	-35%
Mills							
CS7-C-101	0.00035	SS	67,100+	n/a ⁺	n/a ⁺	-90%	n/a ⁺
Filtering centrifuges							
CS7-S-103	0.00002	SS	53,900+	n/a ⁺	n/a ⁺	181%	n/a ⁺
Filters							
CS7-S-104	0.0075	SS	18,600*	n/a ⁺	n/a ⁺	-99%*	-98%
CS7-S-101	1.5	SS	18,600*	n/a ⁺	n/a ⁺	-89%	-47%
CS7-S-102	55	SS	131,000	n/a ⁺	n/a ⁺	n/a ⁺	-37%
CS7-S-105	80	SS	170,200	n/a ⁺	n/a ⁺	n/a ⁺	31%
CS7-S-106	90	SS	184,000	n/a ⁺	n/a ⁺	n/a ⁺	33%
CS7-S-107	90	SS	184,000	n/a ⁺	n/a ⁺	n/a ⁺	33%
Reactor-jacketed agitated							
CS3-R-101	28.4	CS	109,135	n/a ⁺	n/a ⁺	-70%	n/a ⁺
CS3-R-102	28.4	CS	109,135	n/a ⁺	n/a ⁺	-70%	n/a ⁺

* cost of minimum size as size is less than the minimum size

+ not available in this program and so taken from another program

purchase cost not available in EconExpert

TABLE 9. TOTAL MODULE COST (C_{TM}) OF EQUIPMENT

Total module cost	CapCost	DFP	CCEP	EconExpert	AspenPEA
Case study	C_{TM} \$million	$\Delta\%$	$\Delta\%$	$\Delta\%$	$\Delta\%$
1	4.3	-6%	-4%	-36%	-3%
2	1.0	16%	71%	-5%	68%
3	2.5	6%	-11%	-14%	81%
4	4.1	79%	19%	-29%	28%
5	33.3	27%	-23%	-28%	-36%
6	17.0	94%	14%	2%	-11%
7	9.4	34%	7%	8%	14%

and agitated reactors are only available in CapCost and EconExpert. The results obtained from these two are comparable (Table 8).

Total purchase and module cost

We also compared total purchase and total module costs for all pieces of equipment in each case study given by the different programs. For all programs, if a piece of equipment had a size above the valid range, it was divided into multiple units of smaller size and the costing was done by summing C_{TM} for multiple smaller units. If a piece of equipment had a specification below the valid range, then its cost was estimated by taking the lower limit of the valid range. Similarly, if the flowrate was less than the minimum required for costing, the minimum flowrate was used (not actual). Since layers of silver wire gauze inside the reactor cannot be

mapped to any equipment in the programs, it is not evaluated in case four. For similar reasons, the cost of catalyst pellets is also excluded in the total fixed capital cost in case five.

The five programs give comparable purchase and total module costs for the whole plant in each case study (Table 9 and online table VIII), with DFP giving a slightly higher cost

in some cases. In case study two, both the purchase and total module costs obtained by AspenPEA are more than 60% higher than those by CapCost. This is mainly due to the high costs of towers and vessels in AspenPEA. In case three, there is good agreement among different programs except for AspenPEA, which shows higher purchase and total module costs due to the high costs of crystallizers and vessels. In case studies four and six, the high deviations (more than 80%) in both purchase and total module cost in DFP are mainly due to cost differences in vessels and packing towers.

Conclusions

Although based on different methods and developed in different platforms, all five programs are user friendly and are useful tools for estimating the capital cost of chemical process

plants. AspenPEA has the most equipment types available out of the five programs. DFP is the most limited in that regard. Based on our analysis, the overall plant cost does not deviate much among the different programs studied. However, equipment capital costs for different programs may not be comparable. There is generally good agreement for the purchase costs of floating-head heat exchangers. For fixed-head heat exchangers and pumps, there is greater deviation in both purchase and total module costs among different programs studied. Also, there is significant deviation for most vessels and towers in total module cost. Since material, pressure and installation factors vary in different methods in calculating total module cost, capital cost estimates may differ. Hence, while evaluating plant design alternatives, it is important to use only one program for cost evaluation of process design options to maintain consistent results. ■

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Editor's note: Additional tables for total module cost of equipment are included in the online version of this article (www.che.com).

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